

Pooja Kotresh Halannavar

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Profile

A passionate Master's student in Data Science and Artificial Intelligence with strong skills in Data Analysis, Machine Learning and a keen interest in LLM-based automation and building AI solutions. Strong analytical, problem solving, and communication skills, combined with confidence and a drive for continuous learning.

Skills

Programming Languages: Python, Reactjs, Java, HTML, CSS, JavaScript, SQL

Technologies: REST APIs, FastAPI, Docker, MySQL, Git, GitHub, LangChain, VS Code, PyCharm, Linux

Machine Learning Libraries: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, NLTK, Matplotlib, OpenCV, MLflow, Hugging Face Transformers

Languages: English(C1), German(A2.1), Kannada, Hindi

Education

Master's in Data Science and Artificial Intelligence Saarland University, Germany	Oct 2023 – present
Bachelor's in Computer Science (1.72/4) PES Institute Of Technology & Management, India	Aug 2017 – July 2021

Experience

Research Assistant , Deutsches Forschungszentrum für Künstliche Intelligenz, Germany	June 2025 – Present
- Conducted research on compositional reasoning in vision-language models, by evaluating across multiple datasets and analyzing model behavior and failure patterns and communicated research findings. - Built Streamlit interfaces for user data collection and deployed reproducible workflows using Docker.	
Software Engineer , TATA Consultancy Services, India	Oct 2021 – Aug 2023
- Developed reusable components using web technologies (React), and worked on backend part in Python. - Designed automated unit and integration test infrastructure for code quality analysis of the application.	
Full-Stack Development Intern , Srichid Technologies, India	July 2020 – Oct 2020
- Implemented a full stack development project using Angular and Web Api. - Developed the dashboards using HTML, CSS, Javascript and Python.	

Projects

Food Ordering Chatbot

- Built a conversational ordering system integrating Dialogflow ES with a FastAPI backend and MySQL database, enabling users to place and track orders through natural language.
- Designed REST API endpoints to handle intent-driven chatbot requests, process order logic, and manage database communication end-to-end. [Keywords: Dialogflow ES, FastAPI, REST APIs, MySQL, Python]

Violence Detection in Surveillance videos using Deep Learning

- Developed a deep learning-based violence detection system using transfer learning on DenseNet, SlowFast, and I3D models. Fine-tuned on a video dataset with four violence types, achieving 82.14% accuracy with SlowFast. [Keywords: DL, Transfer Learning, 32 frames per video, SVM, FFNN]

Image Caption Generator

- Developed an image captioning model using a pre-trained CNN(VGG16) for feature extraction and LSTM to generate captions word by word. [Keywords: CNN(VGG16), LSTM, BLEU score]

Others: Portfolio, Breast Cancer Prediction using CNN, Handwritten Digits Recognition